

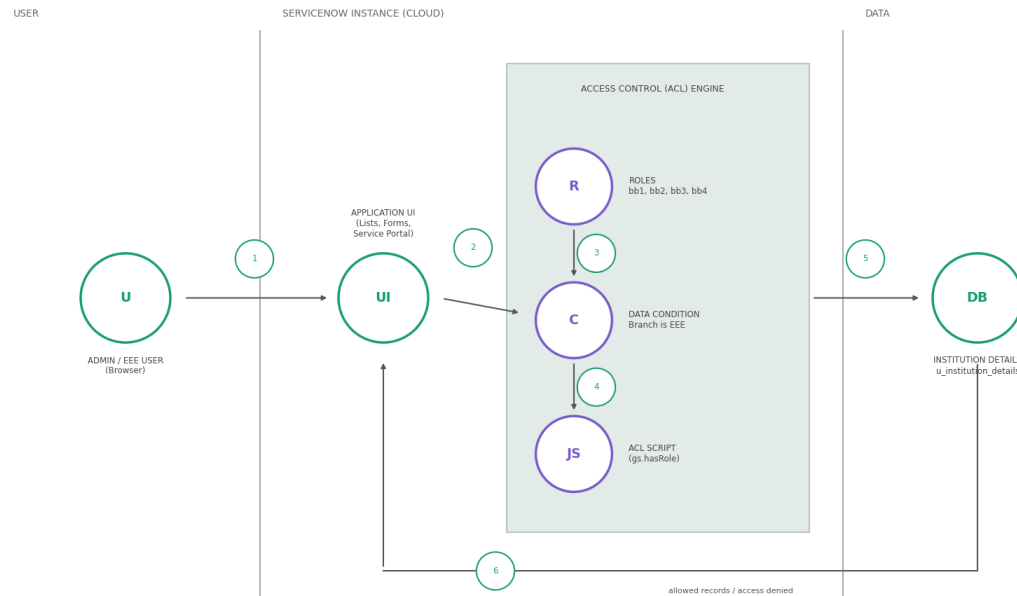
Project Design Phase-II Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	MKU201Team66ab4dee00fc666a751b56cd7
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Script-controlled ACL for restricting record access in ServiceNow



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Admin and EEE users interact with the ServiceNow web UI: Application Navigator, User Administration, lists, forms and Service Portal	ServiceNow Now Platform UI in a web browser, Service Portal
2.	Application Logic-1	Users and roles creation: create the EEE User and the roles bb1, bb2, bb3, bb4, then assign the roles to the user	ServiceNow User Administration (Users, Roles)
3.	Application Logic-2	Table creation: custom table Institution Details (u_institution_details) with its fields and branch records	ServiceNow System Definition (Tables)
4.	Application Logic-3	Script-controlled ACL logic: Read ACL with role bb1, data condition Branch is EEE and a script that allows admin and bb1 users and denies all others; Create, Write and Delete ACLs with roles bb2, bb3 and bb4	ServiceNow Access Control (ACL), JavaScript (gs.hasRole)
5.	Database	Data type and configuration: Student Roll Number (Auto Number), Student Name and Faculty Name (Reference – User), Branch (Choice: ECE, EEE, CSE), Email and Phone Number (String), Description (Multi String)	ServiceNow table u_institution_details
6.	Cloud Database	Database service on cloud: the records are stored in the database of the ServiceNow cloud instance	ServiceNow instance database
7.	File Storage	No separate file storage is required; all data is stored as table records	Not applicable
8.	External API-1	No external API is used in this solution	Not applicable
9.	External API-2	No external API is used in this solution	Not applicable
10.	Machine Learning Model	No machine learning model is used in this solution	Not applicable
11.	Infrastructure (Server / Cloud)	Application deployment on Cloud: the ACLs are configured on a ServiceNow instance using admin access (with the security_admin role elevated) Local Server Configuration: Not required; the instance is accessed through a web browser Cloud Server Configuration: ServiceNow cloud instance	ServiceNow cloud instance

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source framework is used; the solution relies on built-in ServiceNow features	ServiceNow platform features, JavaScript (ACL script)
2.	Security Implementations	Role-based access control: Read, Create, Write and Delete ACLs with roles bb1 to bb4, the data condition Branch is EEE and a script that denies users without the required role; security_admin is elevated to create ACLs; access is tested by impersonating users	ServiceNow ACLs, Roles, security_admin role, gs.hasRole
3.	Scalable Architecture	Access rules are defined at table level, so more branches, roles or operations can be added with extra data conditions, roles or ACLs	ServiceNow table-based ACLs
4.	Availability	The application runs on a cloud-hosted ServiceNow instance and the ACLs are active, so they are checked every time a record is accessed	ServiceNow cloud instance
5.	Performance	The ACL script only checks roles, so access decisions are quick when a list or form is opened	ServiceNow ACL engine, gs.hasRole

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>